

ANSWERS

1.

Question	Answer	Marks
7(a)	 and drawn between T_1 and T_2	B1
7(b)(i)	$(I =) V / R$ or 0.40 / 2.5 0.16 (A)	C1 A1
7(b)(ii)	$(P =) VI$ or 0.40×0.16 or $(P =) I^2 R$ or $0.16^2 \times 2.5$ 0.064 (W)	C1 A1
7(c)	resistance (of the thermistor / circuit) decreases increase in current in circuit / R takes a larger proportion of the total resistance / voltage and voltmeter reading / it increases.	B1 B1

2.

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	$(R =) V / I$ or 1.5 / 0.20 or 7.5 (Ω) $(R_T =) 7.5 - 1.5 - 3.5$ 2.5 (Ω)	C1 C1 A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases or smaller fraction of total resistance	B1
	p.d. across R and S increases or (p.d. is) smaller fraction of e.m.f.	B1

3.

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A) voltmeter V_2 9(.0 V)	B1 B1
6(a)(ii)	$(R =) V / I$ in any form 36 (Ω)	C1 A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference provided the temperature is constant	B1 B1
6(b)	0.35 (A) seen or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$ voltage (across R) 0.35×36 or 12.6 (V) seen 19 V	B1 B1 B1

4.

Question	Answer	Marks
3(a)	($I = $) V/R in any form or (total resistance =) $700\ (\Omega)$	B1
	$0.017\ (A)$	B1
3(b)	($P = $) $I^2 R$ or VI and either V^2/R or $V=IR$ numerical or algebraic in any form	C1
	or voltage across $400\ \Omega$ resistor = (a) $\times 400$ ($6.857\ V$)	A1
	$0.12\ (W)$	
3(c)	circuit drawn with all 3 resistors in parallel with cell / power supply	B1
	ammeter in series with $100\ \Omega$ resistor or in series with resistor with largest current in arrangement drawn	B1

5.

Question	Answer	Marks
5(a)	energy / work done per unit charge or $\frac{\text{energy}}{\text{charge}}$ or $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply or energy used driving charge around a (complete) circuit or energy transferred to electrical energy	B1
5(b)	it lasts for a longer time or more (electrical) energy available or power supply still functions if one cell fails / runs flat or can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases c.a.o.	B1
	current change or voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases c.a.o.	B1
5(c)(ii)	($V_1 = $) $V_0 \times R_1 / R_0$ or $1.5 \times 4000 / (4000 + 8000)$ or $1.5 \times 4000 / 12\ 000$ or $I = V / R$ or $1.5 \times 8000 / (4000 + 8000)$ or $I = 1.5 / 12\ 000$ or $0.125\ \text{mA}$	C1
	$0.50\ V$	A1

6.

Question	Answer	Marks
8(a)	<u>electrons</u> move from the cloth / to the rod	B1
	cloth becomes positively charged	B1
8(b)	mention of <u>free</u> / <u>mobile</u> / <u>delocalised</u> / <u>sea</u> of electrons	B1
	(free) electrons present in metal and not present in plastic	B1
8(d)(i)	(region of space) where an (electrically) charged particle / charge experiences an (extra) force	B1
8(d)(ii)	at least five straight lines outside Y and none curved and perpendicular to the surface	B1
	at least one arrowhead pointing inwards and none wrong	B1
	field lines uniformly spaced at surface of sphere and no crossing / touching anywhere	B1

7.

Question	Answer	Marks
5(a)	middle box of 1st three boxes <u>and</u> top box of 2nd three boxes	B1
5(b)(i)	$(P =) VI$ or I^2R algebraic or numerical seen in any form	C1
	0.0024 W	A1
5(b)(ii)	$V = IR$ algebraic or numerical in any form or potential divider formula seen	C1
	4.8 (V) seen or $6.0 - 1.2 / 0.002$ or $3000 - 600$ or $6 = 0.002 \times (600 + R_V)$ or $1.2 = 6 \times 600 / (R_V + 600)$	C1
	2400 Ω	A1
5(b)(iii)	resistance of thermistor / X increases (on cooling)	B1
	current (in thermistor) decreases or potential divider argument used to explain decrease in pd across Y or pd across X <u>increases</u>	B1
6(a)(i)	top magnet N pole at bottom and S pole at top	B1

8.

Question	Answer	Marks
6(a)(i)	correct symbol 2 or 3 arrows onto resistor symbol	B1
6(a)(ii)	resistance (of LDR) decreases	B1
	(voltmeter reading) increases and more current in circuit or potential divider argument comparing voltage across resistor / LDR / voltage of cell	B1
6(b)(i)	$V = IR$ seen in any form or (voltage across 4 or 5 Ω) 2.4×4.0 seen	B1
	$(I =) 9.6 / 5 = 1.9(2 \text{ A})$	B1
6(b)(ii)	(current in 6.0 $\Omega =$) 4.3(2) (A) or (voltage across 6 $\Omega =$) 25.8 or 25.9(2) (V) seen or resistance of parallel resistors = $20 / 9$ (Ω) or total resistance = $8 \frac{2}{9}$ (Ω)	C1
	35 V	A1

9.

Question	Answer	Marks
5E(a)	$(t =) Q / I$ or $7.2 / 0.0016$ or $7.2 / 1.6$ or 4.5×10^N	C1
	4500 s	A1
5E(b)(i)	does not change and p.d. / voltage across resistor N does not change / ammeter 2 is not in the branch that contains the LDR	B1
5E(b)(ii)	resistance of LDR increases or current (in M / in ammeter 1) decreases	B1
	time taken increases	B1

10.

Question	Answer	Marks
6(a)	1.5 V	B1
6(b)(i)	work done / energy transferred when charge passes through a component	C1
	$\frac{\text{work done / energy transferred}}{\text{charge passed}}$	A1
6(b)(ii)	4.5 – 2.5 or 2.0 (V) seen	C1
	$(R =) V / I$ or $2.5 / (2.0 / 28)$ or $R / 28 = 2.5 / 2.0$	C1
	35 Ω	A1

11.

Question	Answer	Marks
6(a)	work done / energy (transfer) per unit time	B1
6(b)(i)	$(I =) P / V$ in any form numerical or algebraic	C1
	1.7 A	A1
6(b)(ii)	6.9–7.2 Ω	B1
6(c)	resistance (of wire) is larger	B1
	power is less and either current is smaller or $P = V^2 / R$ or $P = VI$ and $V = IR$	B1

12.

Question	Answer	Marks
8(a)	top row 3(.0) V	B1
	potential divider formula / working shown e.g. $1200 / 2000 (\times 6)$ or $800 / 2000 (\times 6)$ or split 6 (V) in ratio 1200:800 or $I = 6 / 2000$	C1
	second row 3.6 (V)	A1
8(b)(i)	correct symbol with arrows <u>towards</u> resistor 	B1
8(b)(ii)	resistance decreases and output voltage increases	B1
8(b)(iii)	thermistor	B1

13.

Question	Answer	Marks
7(a)	lower temperature / cooling of thermistor	B1
	less light on LDR	B1
7(b)(i)	energy / work per unit charge	B1
7(b)(ii)	Any 2 from: - voltage across diode - voltage across ammeter - voltage across internal resistance (of battery) (e.m.f. / voltage of battery =) $5.4 + 0.6$ (or larger)	B2
7(b)(iii)	$(E =) VQ$ or VIt in any form algebraic or numerical	C1
	3200 J	A1
7(b)(iv)	no current / no ammeter reading and diode has been reversed / diode does not conduct / has (very) high resistance / blocks current / reference to current / voltage graph	B1

14.

Question	Answer	Marks
9(a)	correct symbol for lamp	B1
	correct symbol for voltmeter placed across lamp	B1
	correct symbol for variable resistor in series with lamp and battery or used as a potential divider	B1
9(b)	start on 10 A scale (and lower if reading small) or uses 10 A scale for points from 1 to 1.8 A or uses 10 A scale for voltages above 2 V (and for 2 V)	B1
	for lower readings switch to (0-) 1 A scale	B1
9(c)	graph curved or not a straight line (through origin) or current not (directly) proportional to voltage	B1
9(d)(i)	$(R =) V / I$ in any form algebraic or numerical	C1
	at 0.40 V, resistance = $0.80 (\pm 0.03) \Omega$	A1
9(d)(ii)	at 6.0 V, resistance = $3.3\text{--}3.4 \Omega$	B1
9(e)(i)	temperature not constant / increases	C1
	resistance increases as temperature increases	A1
9(e)(ii)	(resistance is directly) proportional to length	B1
	(resistance is) inversely proportional to area or proportional to $1 / \text{area}$	B1
9(e)(iii)	any clear use of proportionality using 0.80Ω	B1
	0.050 m	B1

15.

Question	Answer	Marks
10(a)(i)	(resistance) is <u>directly</u> proportional to length	B1
10(a)(ii)	(resistance) is <u>inversely</u> proportional to (cross-sectional) area	B1
10(b)	$(R =) 6.4 \times 7.5 \times 10^{-4} / 100$ or 4.8×10^{-n} (n is an integer)	C1
	$4.8 \times 10^{-5} \Omega$	A1
10(c)(i)	energy / work done in driving a charge round a circuit or energy transferred to electrical energy	B1
	energy / work done per unit charge or energy / charge	B1
10(c)(ii)	$(R =) 6.4 + 9.6$ or 16	C1
	$(V =) 1.2 \times 6.4 / 16$ or current = $1.2 / 16$ or current = 0.075	C1
	0.48 V	A1
10(c)(iii)	trace moves vertically away from centre line	B1
	moves distance of 2.4 cm	B1
	remains horizontal or distance moved by trace directly proportional to distance moved by jockey or comment about tape	B1
10(d)(i)	the battery lasts twice as long as the single cell	B1
10(d)(ii)	no effect or trace moves in an identical manner	B1
	e.m.f. of identical cells in parallel equals the e.m.f. of a single cell	B1

16.

Question	Answer	Marks
8(a)(i)	resistance decreases as temperature increases	B1
	current increases (and voltage constant) as temperature increases	B1
8(a)(ii)	volume (of liquid/gas), expansion (of solid/liquid or gas), length (of solid), pressure (of gas) length (of bimetallic strip), e.m.f. (of thermocouple), colour (of liquid crystals or very hot objects)	B1
8(b)(i)	<u>temperature</u> of melting ice or freezing water	B1
8(b)(ii)	<u>temperature</u> of boiling water (at 1 atmosphere)	B1
8(c)(i)	work done / energy change $\div$ charge (passed through a component)	B1
8(c)(ii)	$(R =) V / I$ in any form	C1
	$(I =) 0.014(0)$ (A)	C1
	260 Ω	A1
8(c)(iii)	(V across R =) 25-3.6 or 21(.4 V) OR (total resistance =) $25 / 0.014$	C1
	1500 Ω	A1
8(d)(i)	(a thermometer where the quantity) changes a lot per unit temperature rise / $^{\circ}\text{C}$ / same temperature rise	B1
8(d)(ii)	at high temperatures	B1
	large(r) change in current / meter deflection for unit temperature rise / $^{\circ}\text{C}$ / same temperature rise or large(r) gradient	B1
8(e)	heat (thermal energy) produced in the thermistor / resistor	B1

17.

Question	Answer	Marks
8(a)(i)	electrical in first box	B1
	heat	B1
	light	B1
8(a)(ii)	energy output / energy input	C1
	<u>useful</u> output energy / <u>total</u> energy input	A1
8(b)	two decrease	C1
	all three decrease	A1
8(c)(i)	$(R =) V / I$ in any form numerical or algebraic	C1
	total current = 2.5 A or both resistances 24 (Ω) and 6(.0 Ω)	C1
	4.8 Ω	A1
8(c)(ii)	smaller area / diameter / radius	B1

18.

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	$(V =) 12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	$(R =) V / I$ or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

19.

Question	Answer	Marks
6(a)(i)	$(I =) P / V$ or 60 / 240	C1
	0.25 A	A1
6(a)(ii)	$(R =) V / I$ or 240 / 0.25	C1
	960 Ω	A1
6(b)(i)	3.0 V	B1
6(b)(ii)	resistance is smaller	B1
	(filament at a) lower temperature	B1
6(b)(iii)	current surge (due to lower resistance when cold)	B1

20.

Question	Answer	Marks
6(a)	$(I =)V / R$ algebraic or numerical	C1
	0.4(0) A	A1
6(b)	current in Q is 0.25 (A) or (total resistance) $6 / 0.25$ or $6 / 0.65$ or $9.2(3 \Omega)$ seen	C1
	24Ω	A1
6(c)(i)	correct circuit with two lamps ammeter and battery in series	B1
6(c)(ii)	(total) resistance increases or (each) lamp has lower p.d. across it	B1

21.

Question	Answer	Marks
5(a)	adjusts the variable resistor / rheostat or change the number of cells	B1
5(b)(i)	current proportional to voltage or $V = kI$ where k is a constant	M1
	provided temperature / physical conditions unchanged	A1
5(b)(ii)	at least two values of V/I or I/V shown to be the same;	B1
5(b)(iii)	use 0–10 A range for 220 and 320 mA readings and 0–200 mA range for 100 mA reading or start with 10 A / highest range and either ∞ reduce to 200 mA for last reading ∞ reduce range and keeping high value on scale / not over scale	B1

22.

Question	Answer	Marks
10(a)	electrons c.a.o.	M1
	towards the ammeter or away from the negative terminal or towards the positive terminal	A1
10(b)(i)	thermistor c.a.o.	B1
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ or $1 / R_T = 1 / 1.5 + 1 / 6.0$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	$1.2 (\Omega)$	C1
	2.5Ω	A1
10(b)(iii)	$(I =) V / R$ or $12 / 2.5$	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current increases	B1
	voltage (across 1.3Ω) increases	B1
	trace moves towards top of screen / upwards	B1
10(d)(i)	8	B1
10(d)(ii)	1.5 V	B1

23.

Question	Answer	Marks
7(a)(i)	$1/R_T = 1/R_1 + 1/R_2$ or $1/R_T = 1/3.6 + 1/1.8$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.8 \times 3.6 / (1.8 + 3.6)$	C1
	1.2 (Ω)	C1
	4.0 Ω	A1
7(a)(ii)	$(I =) V/R$ or 6.0 / 4.0	C1
	1.5 (A)	A1
7(b)(i)	work done / energy (released) per <u>unit</u> charge (passed through component)	B1
7(b)(ii)	voltmeter symbol and across 2.8 Ω resistor	B1
7(b)(iii)	<u>total</u> resistance increases or resistance of <u>circuit</u> / <u>parallel combination</u> increases	B1
	current (in 2.8 Ω resistor) decreases	B1

24.

Question	Answer	Marks
10(a)	work done / energy transferred	B1
	work done / energy transferred in taking unit charge (through resistor)	B1
10(b)(i)	$(I =) V/R$ in any form algebraic or numerical	C1
	0.24 A	A1
10(b)(ii)	(total resistance of) 50 (Ω) seen or 2.4 (V) p.d. across 10 Ω resistor	C1
	12 V	A1
10(c)	voltmeter chosen is 0–20 V	B1
	correct explanation of why 0–200 V meter is unsuitable OR correct explanation of why 0–2 V meter is unsuitable	B1
10(d)(i)	$(P) = VI$ or $I^2 R$ or V^2 / R in any form	C1
	0.58 W	A1
10(d)(ii)	$\frac{1}{2}P$ rated resistor blows / too hot / melts	B1
10(e)	current in 10 Ω increases <u>resistance</u> decreases (of parallel combination or total resistance)	B1
	p.d. across 10 Ω increases p.d. is proportional to current or (p.d.) increases as current increases or potential divider argument	M1+A1
	p.d. across 40 Ω decreases sum of p.d.s constant / 12 V / value in (b)(ii) or p.d. across 10 Ω increases or resistance (40 Ω and R) a smaller fraction of total	B1

25.

Question	Answer	Marks
5(a)	correct circuit symbols for battery, resistor, <u>variable resistor</u> , ammeter and voltmeter	B1
	ammeter in series with resistor and voltmeter in parallel in correct series circuit	B1
5(b)(i)	$(R =) V / I$ in any form numerical or algebraic	C1
	3.5 Ω	A1
5(b)(ii)	voltage or current or power too large	B1
	calculation of power as 5(.04) W or 5.6 W or max voltage as 3.4(2) V or 3.2(4) V or max current as 0.88 A or 0.93 A	B1
5(b)(iii)	no and half voltage (2.1 V across each) or half current (0.54 A through each) or quarter power (1.13 W in each)	B1

26.

Question	Answer	Marks
7(a)(i)	$1/R = 1/R_1 + 1/R_2$ or $R_1 R_2 / (R_1 + R_2)$ or $1/R = 1/1800 + 1/9000$ or $9000 \times 1800 / 10\ 800$ or 0.00066667	C1
	1500 Ω	A1
7(a)(ii)	$(I =) V / R$ or 4.5 / 1500	C1
	0.0030 A or 3.0×10^{-3} A or 3.0 mA	A1
7(b)(i)	increases and resistance of LDR decreases	B1
7(b)(ii)	does not change and resistance / e.m.f. does not change / not affected by LDR	B1

27.

11(b)(i)	$1/R = 1/R_1 + 1/R_2$ or $R_1 R_2 / (R_1 + R_2)$ or $1/R = 1/15 + 1/60$ or $15 \times 60 / 75$ or $15 \times 60 / (15 + 60)$	C1
	12 (Ω) or 0.083 (Ω)	C1
11(b)(ii)	30 Ω	A1
	$(I =) V / R$ or 7.5 / 30	C1
11(b)(iii)	0.25 A	A1
	18 Ω resistor underlined 2 and 60 Ω resistor underlined	B1
11(b)(iv)	five cells in series and all in same direction	B1
11(b)(v)	resistance increases and current decreases	B1

28.

Question	Answer	Marks
8(a)	equal (numbers of) positive and negative charges	B1
8(b)(i)	negative charge moves from cloth to rod	C1
	electrons move from cloth to rod	A1
8(b)(ii)	apparatus needed, e.g. (small) pieces of paper / water stream / (gold leaf) electroscope / suspended or pivoted other <u>charged</u> rod / charged object / conducting object	B1
	correct statement of what is seen / felt with apparatus	B1

29.

Question	Answer	Marks
10(a)(i)	directly proportional	B1
10(a)(ii)	straight line or does not curve or constant gradient	B1
10(a)(iii)1	greater or twice as large	B1
10(a)(iii)2	straight line with half the gradient	B1
10(b)(i)1	$1/R_t = 1/R_1 + 1/R_2$ in any form e.g. $1/R = 1/20 + 1/80$ or 16 (Ω) seen	C1
	40 Ω	A1
10(b)(i)2	$(I=)V/R$ in any form e.g. 6 / 40	C1
	0.15 A	A1
10(b)(i)3	0.15 $\times$ 16 or 0.15 $\times$ 24 or 3.6 (V) seen or current split in ratio 1:4, e.g. 0.03 A and 0.12 A seen or clear attempt at potential divider formula	C1
	2.4 V	A1
10(b)(ii)1	work done $\div$ charge	B1
10(b)(ii)2	correct circuit symbol for a cell and positive correct	B1
	four cells, correct symbol, correctly in series	B1
10(b)(ii)3	four cells in series and another four in parallel or any other series and parallel arrangement of 8 cells with connections to and from battery	B1
10(c)	lasts longer or if one cell fails it still works or contains more energy	B1

30.

Question	Answer	Marks
5(a)	$1/R_t = 1/R_1 + 1/R_2$ or $R_t = R_1 R_2 / (R_1 + R_2)$ formulae or numbers using 20 and 40 Ω or 13(.3 Ω)	C1
	$R_t = R_1 + R_2$ or 20 + any attempt at parallel calculation	C1
	33(.3) Ω	A1
5(b)	$V_2 = V_3$	B1
	V_1 largest or larger than either V_2 or V_3	B1

31.

Question	Answer	Marks
11(a)	battery and ammeter symbols correct	B1
	thermistor symbol correct	B1
	all connected in series	B1
11(b)(i)1	0.05 (A) seen	C1
	$(V =) IR$ or 0.05×240	C1
	12 V	A1
11(b)(i)2	$(R =) V / I$ or 0.12 (A) seen	C1
	100 Ω	A1
11(b)(ii)	description of change in resistance in two equal changes in temperature, e.g. 240–100 and 100–48	C1
	difference values obtained are not the same, with one value shown e.g. 140 and 52 (Ω)	A1
11(c)(i)	two different metals connected together	B1
	voltmeter, ammeter or galvanometer in series	B1
	clear junction of two different metals used as the place to measure temperature	B1
11(c)(ii)	ANY 2 lines from measures high(er) temperatures more sensitive to changes in temperature measures fast(er) changing temperatures / reading can be taken more rapidly produce electronic output / output may be stored or processed / may be read remotely	B2

32.

- (a) (amount of) energy/work (dissipated by source)
by unit charge (around a circuit) M1
A1
- (b) (i) 1 they are the same B1
or $I_B = I_1 = I_2$
- 2 $E = V_1 + V_2$ B1
- (ii) $(I =) V / R$ in any form algebraic or numerical C1
0.25 A A1
- (iii) 4.5 V B1
- (iv) $(P =) VI$ or $(P =) I^2 R$ or $(P =) V^2 / R$ C1
in any form algebraic or numerical
1.1(25) W A1

(c)	current is (directly) proportional to voltage or voltage / current is a constant law holds for constant physical conditions/ constant temperature / constant pressure / for metals	B1 B1
(d) (i)	(directly) proportional or $(R) \propto 1$	B1
(ii)	inversely proportional or $(R) \propto 1/A$	B1
33.		
(a) (i)	same/equal or $I_B = I_1 = I_2$	B1
(ii)	(p.d. of) battery is sum of (p.d. across) fixed resistor and (p.d. across) the variable resistor or $V_B = V_1 + V_2$	B1
(b)	(I=) V/R numerical or algebraic in any form 0.006(0)A	C1 A1
34.		
(a) (i)	$(I =)V/R$ or $12/(6000 + 2000)$ or $12/8000$ or $12/2000$ or $12/6000$ or in (ii) $(V =)IR$ or 0.0015×6000 1.5mA	C1 A1
(ii)	9.0V	B1
(b)	(reading / it) increases resistance of LDR falls	B1 B1
(c)	light meter / sensor or automatic light switch or something sensible	B1
35.		
(a) (i)	$(I =)V/R$ or $6/60$ 0.1(0)A	C1 A1
(ii)	$(I =)P/V$ or $0.9/6$ or 0.15 (A) seen 0.25A	C1 A1
(b) (i)	lamp correctly drawn in series with resistor but not the lamp	B1
(ii)	less voltage (across lamp) because some voltage across resistor / shares voltage with resistor or less current because of effect of resistor	B1

- 36.
- (a) (i) lasts longer **or** one cell can be replaced without switching off the circuit **or** less (internal) resistance **or** if one fails the others still work B1
- (ii) 1.5 V B1
- (b) (i) $(R =)V/I$ **or** $1.5/0.075$ C1
 $20\ (\Omega)$ **or** $1.5/(0.075 - 6.0)$ C1
 $14\ \Omega$ A1
- (ii) decreases B1
 resistance of wire increases B1
- 37.
- (a) (i) energy to drive charge around a circuit **or** terminal p.d. on open circuit B1
 energy to drive **unit** charge around a circuit **or** energy/charge B1
- (ii) lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (not greater e.m.f./voltage) B1
- (b) (i) $4.0\ \Omega$ B1
- (ii) $(1/R_{\text{tot}} =)1/R_1 + 1/R_2$ **or** $1/3 + 1/X$ **or** product/sum **or** $(3 \times X)/(3 + X)$ **or**
 $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$ C1
 $6.0\ \Omega$ A1
- (c) (i) $(I =)V/R$ **or** $2.0/4.0$ C1
 $0.50\ \text{A}$ A1
- (ii) (from) 0 **and** (to) 0.50 to 5.0 A B1
- (d) $I_2 = I_3 + I_X$ B1
- (e) (i) 1.0 V B1
- (ii) 1.0 V B1
- (f) (i) temperature decreases B1
 resistance decreases B1
- (ii) greater than 0.75 A (e.c.f. resistance increases in (f)(i)) B1

38.

- (a) ammeter and voltmeter correct symbols [B1]
 ammeter in series with lamp [B1]
 voltmeter in parallel with lamp [B1]
- (b) R limits or reduces the current/voltage [B1]
 otherwise lamp blows
 or more of the $50\ \Omega$ can be used to adjust voltage / current [B1]
- (c) (i) 12 V, 0.25 A correctly plotted (by eye) [B1]
 curved line from origin [B1]
 correct curvature – decreasing slope [B1]
- (ii) straight line (for fixed resistor) [B1]
lamp has changing temperature or changing resistance
 or fixed resistor has constant temperature or constant **resistance** [B1]
- (d) (i) ($I =$) V/R in any algebraic or numerical form, e.g. 12/50 [C1]
 0.24 A [A1]
- (ii) 0.49 A [B1]
- (iii) 6(.0) V [B1]
- (iv) 12(.24) Ω [B1]

39.

- (a)

A	B
6(.0) V	0 (V)

 [B1, B1]
- (b) (i) resistance (of thermistor) decreases [B1]
 current increases **and** larger voltage across $2000\ \Omega$ /fixed resistor
 or smaller **fraction** of voltage across thermistor **or** potential divider explained [B1]
- (ii) ($I =$) V/R in symbols or numbers [A1]
 0.002(0) A; 2 mA [C1]

40.

- (a) (i) work done **or** energy produced/needed [M1]
per unit charge **or** per coulomb (passing through lamp) [A1]
- (ii) 1. not straight **or** curves **or** gradient changes **or** data used correctly [B1]
2. gets hotter **or** temperature changes **or** resistance increases [B1]
- (iii) 1. 350 mA **or** 0.35 A [B1]
2. $Q = It$ in any form algebraic or numerical **or** $2 \times 60 \times 60$ **or** 7200 (s) seen [C1]
2520 C **or** 2500 C [A1]
3. ($E =$) QV or VIt in any algebraic or numerical form, e.g. $0.35 \times 6 \times 2$ [C1]
15100 J **or** 15120 J **or** 15000 J **or** 4.2 kWh [A1]
4. current and/or voltage falls/varies
or some energy remains (in cell)
or some energy/heat produced **in cell**
or correct argument involving internal resistance of cell [B1]
- (b) lasts longer [M1]
(because) larger energy (initially) **or** smaller current (in each cell) [A1]
or
avoids failure if one cell fails (M1)
(because) other (parallel) cell takes over (A1)
or
larger current/power (in external resistor/lamp) (M1)
(because) smaller (internal) resistance of combined cells (A1)
- (c) symbol for LED clear and in correct direction for cells shown [B1]
4 cells (not all in parallel or in series), switch and LED (labelled or shown) that
switches on and off properly [B1]
cells connected correctly to give 3.0 V total e.m.f., e.g. two sets of parallel cells
connected in series or three in parallel and one in series [B1]

41.

- (a) work done per (unit) charge/coulomb/C **or** energy transformed per (unit) charge/coulomb/C B1
property of a source (of electricity) **or** energy transformed to electrical energy per (unit) charge/coulomb/C B1
- (b) (i) ammeter in series B1
(ii) voltmeter in parallel with lamp **or** lamp and ammeter B1
- (c) (i) ($V =$) 2.0 (V) C1
($R =$) V/I **or** $2.0/0.70$ C1
 $2.9/2.86 \Omega$ (i.e. 2 **or** 3 s.f. only) A1
(ii) (resistance) increases B1
- (d) ($P =$) VI **or** ($P =$) V^2/R **or** $I^2 R$ **or** 12×2.0 **or** 12×0.70 C1
24 W A1
- (e) (i) emission of electrons M1
from heated metal/named metal/filament/wire A1
(ii) 1. prevents collision with air (molecules) **or** prevents deflection **or** lets electrons/particles reach screen/travel unimpeded B1
2. moves vertically (e.g. up/down/above/below **or** vertical line) **not** with horizontal movement due to this voltage B1
attracted by positive **or** repelled by negative **or** attracted by one plate and repelled by the other **or** electric field (acts on charge) B1

42.

- (a) (i) ($I =$) V/R **or** $6.0/12.0$ **or** $6.0/(4.0+8.0)$ **or** (in (ii)) ($V =$) IR **or** 0.50×4.0 C1
0.50 A A1
(ii) 2.0 V (scores C1 in (a)(i) if not already scored) A1
- (b) (i) increased **or** becomes 1.25 A B1
(ii) decreases **or** becomes 0.8Ω B1
- (c) moves up or down **or** $5.0/2.0$ C1
moves up or down by 2.5 cm A1

- 43.
- (a) (i) 2 to 2.1 (V) to any value between 11 and 12 (V)
or above 2/2.1(V) B1
- (ii) temperature increases / gets hotter B1
- (b) (i) (rate of) flow of charge/electrons B1
- (ii) 0.35 A cao B1
- (iii) $(I =) V/R$ algebraic C1
or $6/20$
or $6/0.35$
- 0.3(0) (A) C1
or $1/R_T = 1/20 + 1/17.1$
or $(R_T =) 9.2 (\Omega)$ seen
- 0.65 A A1
- 44.
- (a) current is directly proportional to voltage (**accept** voltage/current = constant, but
not just = R) B1
if temperature/physical conditions constant B1
- (b) $(R =) V/I$ in any form algebraic or using any value of V and I from graph C1
 20Ω A1
- (c) (i) 40Ω or $2 \times (b)$ B1
- (ii) straight line graph through origin below given line ecf (b) (e.g. **accept** above
line if $R < 20$) M1
goes through 0.1 A at 4 V ecf (b) (e.g. allow through 0.2 A at 2 V if $R = 10 \Omega$) A1
- 45.
- (a) correct symbol for thermistor B1
in series circuit with any power supply (e.g. cell or two circles) and a fixed resistor B1
- (b) (i) 12 (V) B1
0.018 (A) B1
- (ii) $(P =) VI$ in any form algebraic or numerical with any voltage (4, 8 or 12) C1
0.14(4) W A1
- (iii) above maximum power or gets too hot or blows up or fails B1

- 46.
- (a) (i) $(V =)IR$ or 0.025×600 C1
15 V A1
- (ii) $5(V)$ or $5/0.025$ or 800 or 800–600 C1
200 Ω A1
- (b) (i) decreases B1
- (ii) ammeter: opposite to (i) B1
voltmeter: same as ammeter (both changes correct)
- 47.
- (a) three lines from one sphere to the other **and** some lines should spread out as they leave one sphere **and** come together nearing the other B1
- correct direction on at least one line and none wrong B1
- (b) $(I =) Q/t$ in any form numerical or algebraic C1
 $2.4 \times 10^{-3} \text{ A}$ A1
- 48.
- (a) (i) **correct circuit symbols** containing, in any circuit, a
- battery/cell/d.c. power supply
 - ammeter
 - voltmeter
 - fixed resistor
- B1
- ammeter clearly measures current through W B1
- voltmeter clearly across W if W shown or a resistor if not B1
- (ii) Any 2 from
- resistance (calculated from) V/I or $V = IR$ seen
 - length (of wire), V and I **all three** measured
 - change length **and** V and I measured
- B2
- (iii) 1. resistance/resistivity changes (with temperature)
OR wire gets hot **and** melts/burns/catches fire/dangerous
OR V only proportional to I at constant temperature B1
2. use of a water bath/heat sink
OR use small currents
OR take reading (quickly) and switch off B1

(b) (i) $(V =) IR$ in any form numerical or algebraic 2(.0)V	C1 A1
(ii) 0.1(0)A	B1
(iii) (Z) has the same potential difference/voltage	B1
(Z) has less/small(er) current (thus larger resistance)	B1
(iv) (p.d. across X =) 0.3×10 (V) OR $(R_Z =) 2/0.1$ OR 20 (Ω) seen	C1
(total p.d.) 5 (V) OR 6.7 (Ω) seen OR $1/R_T = 1/R_1 + 1/R_2$ in any form numerical or algebraic OR 20/3 seen	C1
16.7 Ω ; 17 Ω ; 16.67 Ω ; 16.66 Ω	A1
49.	
(a) LED; light-emitting diode	B1
(b) energy/work done per unit charge/coulomb	B1
(c) voltage/p.d. across P one cell connected the wrong way/acting against the others	B1 B1
(d) (i) correct arrangement	B1
(ii) cells last longer; cells run down slower; one cells fails the others still work; reduces (internal) resistance; if cell removed circuit not broken	B1
50.	
(a) (i) power supply, (wire/resistor/bulb) and ammeter in series voltmeter across wire/resistor/bulb labelled/clear variable power supply or rheostat in series or potentiometer correct symbols or labelled throughout	B1 B1 B1
(ii) read ammeter and voltmeter / measure voltage and current vary power supply/rheostat/current	B1 B1
(iii) $(R =) V/I$ (ign. V/A)	B1
(b) horizontal line and above axis	B1

- 51.
- (a) ammeter in series with supply // ammeters in series with A **and** in series with B & C B1
 A across cell with no switch (condone closed switch) **not**  B1
 B and C in series with switch (closed or open) and cell B1
- (b) (i) ($R =$) V/I in any form numerical or algebraic, e.g. $8/50$, $8/0.05$ C1
 $160\ \Omega$ A1
- (ii) $50\ \text{mA}$ // $0.05(0)\ \text{A}$ B1
- 52.
- (a) circuit diagram with cell and ammeter in series with resistor/wire/lamp and B1
 voltmeter across resistor/wire/lamp B1
 $R = V/I$ in any form or gradient of V, I graph B1
 thermometer/thermocouple used or shown B1
 statement of how different temperatures obtained,
 e.g. water bath/oven/heat room/change supply voltage or current or series resistor
 to change temperature B1
- (b) (i) resistance increases with temperature M1
 (approximately) linear, proportional, straight line increase A1
- (ii) curved line starting at origin C1
 correct curvature from origin with decreasing gradient
 allow zero gradient not negative gradient A1
- (c) (i) 1 (current) increases B1
 thermistor resistance decreases B1
- 2 (voltmeter reading) increases B1
 greater fraction of voltage across resistor or potential divider equation
 explained
 or greater current through fixed/constant/ $2000\ \Omega$ resistor B1
- (ii) (voltage across thermistor) $2.2\ (\text{V})$ or attempt to use potential divider formula C1
 (current) $3.8 / 2000$ or $1.9 \times 10^{-3}\ (\text{A})$
 or $3.8 = 6 \times 2000 / (R + 2000)$ or other correct potential divider equation C1
 $1200\ \Omega$ allow $1157 - 1160$ A1

- 53.**
- (a) variable resistor **or** rheostat B1
- (b) curved line starting at origin **allow** straight at first **not** two straight lines C1
correct curvature **from origin** with decreasing gradient A1
- (c) (i) (resistance) increases (as p.d. increases) B1
- (ii) explanation, e.g. lower current than expected for given p.d. **or** given current needs larger p.d. **or** correct explanation involving $R = V/I$ **accept** gradient decreases B1
- 54.**
- (a) 12(.0) V B1
- (b) top row: 4.6 **and** 0 B1
bottom row: square 1 = square 2 + square 3 **or** 9.2 B1
bottom row: 4.6 in squares 2 **and** 3 **cao** B1
- (c) ($E=$) QV **or** VIt **or** 200×12 C1
2400 J **accept** 2370–2410 J e.c.f. A1
- 55.**
- (a) $R = V/I$ in any algebraic (e.g. $V = IR$) or numerical form C1
1200 Ω A1
- (b) decreases M1
to constant value/to 0.2 A A1
- (c) longer **or** thinner **or** hotter **or** material/made of poorer conductor (higher resistivity) B2
- 56.**
- (a) (i) 6 Ω B1
- (ii) $1/R = 1/R_1 + 1/R_2$ algebraic or numerical C1
2 Ω A1
- (b) $I = V/R$ algebraic or numerical C1
6 A ecf (ii) A1
- (c) ($I =$) 2 (A) **or** proportionality idea/potential divider idea seen C1
8 V A1

57.		
(a)	0 (V)	B1
(b)	(i) 8 Ω (i.e. accept 1 sig.fig.)	B1
	(ii) $R = V/I$ any algebraic form in (ii) or (iii)	B1
	2 A (i.e. accept 1 sig.fig.) ecf (i)	B1
	(iii) 16/8 in (ii) or (ii) $\times$ 6	C1
	12 V ecf (ii)	A1
58.		
(a)	arrow anticlockwise anywhere near top line of circuit	B1
(b)	LDR or light dependent resistor	B1
(c)	less resistance of X	B1
	same change in voltage as resistance	
	(voltage decreases alone B1)	B1
59.		
(a)	4.5 V	B1
(b)	$I = V/R$ in any form using symbols or words	B1
	4.5/15	C1
	0.3 A	A1
(c)	provides smaller (internal) resistance or lasts longer or less lost voltage or one (cell) fails others work or less heat/energy lost	B1
60.		
(a)	(i) voltage past maximum or 3V / off scale / outside range	B1
	(ii) reading less accurate or sensitive / not far up scale or smaller deflection	
(b)	(i) $V = I R$ in any algebraic format	B1
	4/12	C1
	0.33 A (accept 1/3 A)	A1
	(ii) (i) $\times$ 30 or (i) $\times$ 18 + 4 or $30 \times 4/12$	C1
	9.9 – 10 V (e.c.f (i), e.g. if (i) = 0.3, $0.3 \times 30 = 9V$ or $0.3 \times 18 + 4 = 9.4 V$)	A1
	only 1 unit error in this question	